Secure Control and Management Plane (SCMP): From AI Racks to Autonomous Infrastructure

Ford Tamer



Sanjoy Maity



OCP GLOBAL SUMMIT

October 12-15, 2026
San Jose, California

Our Unprecedented Opportunity for Collaboration

OPEN Compute Project

RAPID SCALE

Data Center AI Capex

$1.2T → **$10T**

2027[1] 2030[2]

Semi Industry

$400B

Q2'26 Actual Revenue[3]

Physical AI TAM

$50T

Forecast[4]

AUTONOMY

From Humanoids to Dark Data Centers



NEXT FRONTIER

Space Data Centers



Sources: 1. Morgan Stanley, Sept 2026; 2. SemiAnalysis, August 25, 2026, 3. JP Morgan, August 2026, 4. Jensen Huang, GTC March 2026

LATTICE SEMICONDUCTOR

Four Challenges Driving Infrastructure Decisions

OPEN Compute Project

ACCELERATED
SHIFT LEFT

INCREASING
COMPLEXITY

DISTRIBUTED
SECURITY

FUTURE-PROOFING
INFRASTRUCTURE

SOLUTION:
SECURE CONTROL AND MANAGEMENT PLATFORM

LATTICE SEMICONDUCTOR

Four Challenges Driving Infrastructure Decisions

OPEN Compute Project

ACCELERATED
SHIFT LEFT
OPF/OSF, OCPDiag

INCREASING
COMPLEXITY
DC-MHS

DISTRIBUTED
SECURITY
OpenPROT, OCP S.A.F.E.™

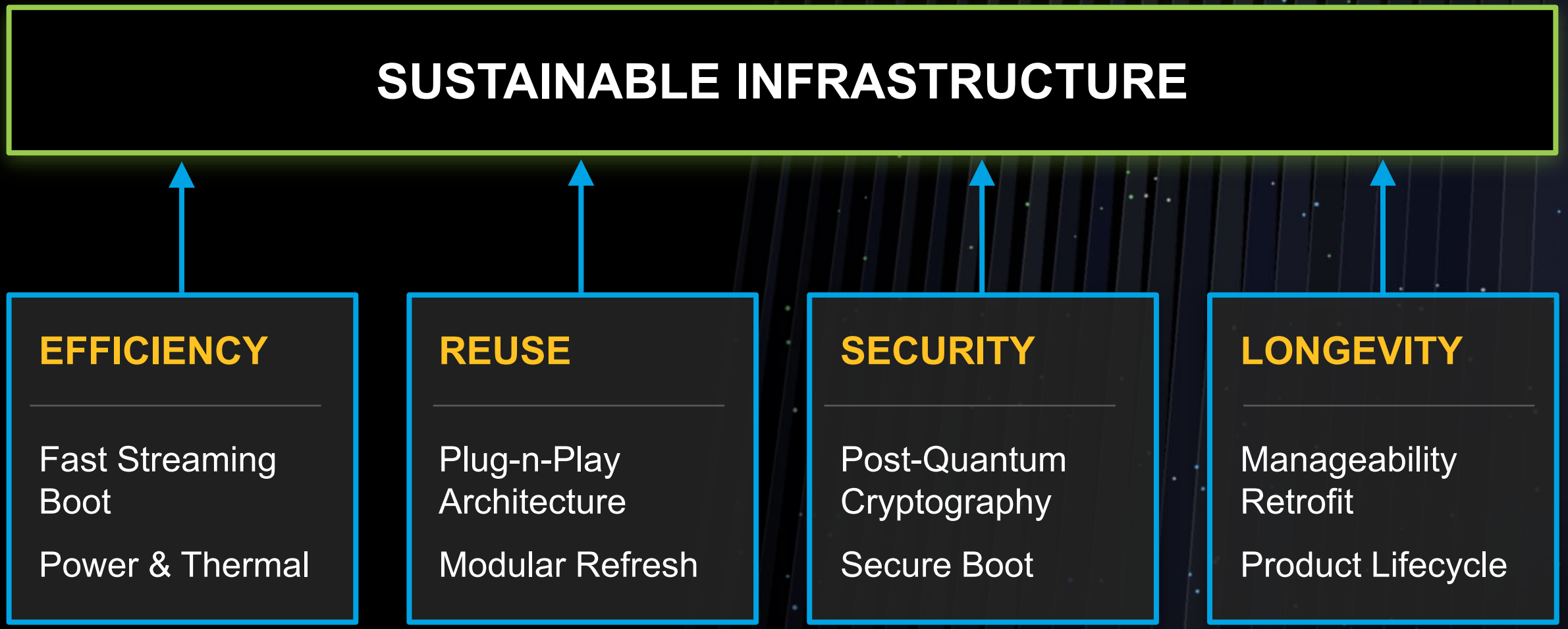
FUTURE-PROOFING
INFRASTRUCTURE
OpenRMC, ORV3

OCP Initiatives

FOUNDATION:
SUSTAINABLE INFRASTRUCTURE

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Sustainability is Mandatory for an Autonomous Future



OCP Has Built a Strong Open Foundation

OPEN Compute Project

INITIATIVES

- Hardware Management / OpenRMC-DM
- Security / S.A.F.E.™
- MHS / DC-SCM
- Rack & Power

OPEN FIRMWARE ECOSYSTEM

OpenBMC

Caliptra

tianocore

- EDK II
- OpenBMC
- Caliptra™ / Open PRoT

MANAGEMENT STANDARDS

DMTF

- Redfish®
- MCTP / PLDM
- SPDM

The foundations exist. The opportunity is to connect them.

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Lattice + AMI Existing OCP Participation and Contributions

Workgroups / Workstreams

DC-MHS / DC-MHS – mPnP

DC-SCM / LTPI

OBMF-ICP

OpenRMC (OpenRMC-DM)

HPC

OpenChiplet / FCSA

OPF / OpenSFI

OPF / UPCI

Firmware Integrity and Manifest

Security / OpenPRoT

Source Code

Aptio Community Edition

MegaRAC OneTree Community Edition

Tektagon Community Edition

Marketplace

Aptio Community Edition

LTPI IP Core

MegaRAC OneTree Community Edition

MPESTI IP Core

Universal Backplane Management Controller

Orchestrating for the Autonomous Rack Infrastructure

NEW OCP WORKSTREAM LAUNCHING Q4'26

Secure Control and Management Plane

Discover | Verify | Coordinate | Recover | Validate

FOUNDING CONTRIBUTORS

ami (A LATTICE COMPANY) · LATTICE SEMICONDUCTOR · AMD · arm · ASPEED · intel · JABIL · Microsoft · NXP

ECOSYSTEM PARTICIPANTS

Operators | ODMs / OEMs | Silicon Providers | Firmware Providers | Power & Cooling | Open-source Communities

INDUSTRY FOUNDATIONS

OPEN Compute Project · OpenBMC · Caliptra · tianocore · DMTF

LATTICE SEMICONDUCTOR

Bringing Leading Control & Management to the Data Center

OPEN Compute Project

LATTICE SEMICONDUCTOR

FPGA



Low Power,
Deterministic,
Programmable



ami™
A LATTICE COMPANY

PLATFORM FIRMWARE



Boot,
Manageability,
Security

Delivering Secure Autonomous Operation from Node to Rack to Pod to Grid

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Lattice and AMI Neutral as Companion Chip and Firmware

OPEN Compute Project

Data Center AI

Physical AI

GPU/TPU AI Accel.	CPU	Switch, NIC	MPU/MCU	BMC	Sensor	Analog
NVIDIA	intel	NVIDIA	NXP	ASPEED	SONY	ANALOG DEVICES
AMD	AMD	BROADCOM	ST	nuvoTon	OMNIVISION	TEXAS INSTRUMENTS
Google	arm	MARVELL	MEDIATEK	NXP	HESAI	Infineon
amazon	NVIDIA	AMD	Qualcomm	HPE	SICK	ROHM SEMICONDUCTOR
Microsoft		REALTEK	RENESAS	DELL	SAMSUNG	
Meta		cisco		AXIADO	Valeo	
Alibaba Cloud				StarFive	TELEDYNE FLIR	
ByteDance					MAGNA	
					e-con Systems Think Camera. Think e-con.	

Data Plane Processing | Control Plane | Physical World

Secure Control and Management Platform

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Four New AMI + Lattice Solutions Already Launched

OPEN Compute Project

SECURITY RETROFIT

PLUG-AND-PLAY

Secure Control and Management Plane

RACK BOOT

POWER + COOLING

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OCP COMMUNITY CALL TO ACTION

Secure Control and Management Plane

Join the SCMP Workstream

Define the Architecture

Align the Interfaces

Contribute a Reference Design

FOUNDING CONTRIBUTORS

AMD
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arm
ASPEED
intel
JABIL
LATTICE SEMICONDUCTOR
Microsoft
NXP

The Future is Autonomous

Open secure control and management is how we get there



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